Naïve Bayesian

Tennis Playing

- Tennis playing record

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Tennis Playing

- Today is Sunny, Mild, High and Strong
  - Will he play tennis today?

- How can you predict his tennis playing?
  - Let's use Probability

$$P(yes \mid sunny, mild, high, strong)$$

$$P(no \mid sunny, mild, high, strong)$$

Tennis Playing

- Naïve Bayesian Classifier
  - Probability with the strongest assumption on independence

If Output is given,
Inputs are independent from each other.

$$P(yes \mid sunny, mild, high, strong) = \frac{P(sunny, mild, high, strong \mid yes)P(yes)}{P(sunny, mild, high, strong)}$$

$$P(no \mid sunny, mild, high, strong) = \frac{P(sunny, mild, high, strong \mid no)P(no)}{P(sunny, mild, high, strong)}$$

Tennis Playing

- Naïve Bayesian Classifier

$$P(yes), P(no)$$

$$P(sunny, mild, high, strong \mid yes)$$
$$= P(sunny \mid yes)P(mild \mid yes)P(high \mid yes)P(strong \mid yes)$$

$$P(sunny, mild, high, strong \mid no)$$
$$= P(sunny \mid no)P(mild \mid no)P(high \mid no)P(strong \mid no)$$

By assumption of indepenence

$$P(sunny, mild, high, strong) = ??$$

Do we need to evaluate this?
Can we evaluate this?

Tennis Playing

- Naïve Bayesian Classifier
  - Let's ESTIMATE the followings, but HOW?

$P(yes)$,
$P(no)$,
$P(sunny \mid yes)$,
$P(mild \mid yes)$,
$P(high \mid yes)$,
$P(strong \mid yes)$,
$P(sunny \mid no)$,
$P(mild \mid no)$,
$P(high \mid no)$,
$P(strong \mid no)$

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
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9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Tennis Playing

- Naïve Bayesian Classifier

$$P(yes \mid sunny, mild, high, strong)$$
$$= \frac{P(yes)P(sunny \mid yes)P(mild \mid yes)P(high \mid yes)P(strong \mid yes)}{P(sunny, mild, high, strong)}$$
$$= \frac{1}{\alpha}\left(\frac{9}{14} \times \frac{2}{9} \times \frac{4}{9} \times \frac{3}{9} \times \frac{3}{9}\right) = \frac{0.007055}{\alpha}$$

$P(sunny, mild, high, strong) = \alpha$

$$P(no \mid sunny, mild, high, strong)$$
$$= \frac{P(no)P(sunny \mid no)P(mild \mid no)P(high \mid no)P(strong \mid no)}{P(sunny, mild, high, strong)}$$
$$= \frac{1}{\alpha}\left(\frac{5}{14} \times \frac{3}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{3}{5}\right) = \frac{0.04114}{\alpha}$$

Will he play or not?

Tennis Playing

- Naïve Bayesian Classifier
  - Can we evaluate $P(sunny, mild, high, strong)$

$$P(yes \mid sunny, mild, high, strong) + P(no \mid sunny, mild, high, strong) = 1$$

$$= \frac{0.007055}{\alpha} + \frac{0.04114}{\alpha} = 1$$

Why?

$$\alpha = P(sunny, mild, high, strong) = 0.048195$$

Formal Description of NBC

- Assume target function $f: X \rightarrow V$,
- What is the most probable value of f(x) where x is $(a_1, a_2 .. a_n)$
  - Let's estimate as follows:

$$f(x) = \arg\max_{v \in V} P(v \mid a_1, a_2, \cdots, a_n)$$

  - What does it mean?

  - It is equivalent

$$f(x) = \arg\max_{v \in V} \frac{P(a_1, a_2, \cdots, a_n \mid v) P(v)}{P(a_1, a_2, \cdots, a_n)}$$

$$= \arg\max_{v \in V} P(a_1, a_2, \cdots, a_n \mid v) P(v)$$

Formal Description of NBC

- However, we don't know the values of

$$P(a_1, a_2, \cdots, a_n \mid v), P(v)$$

- Let's estimate those with GIVEN DATA
  - Still, we may have enough data to estimate $P(a_1, a_2, \cdots, a_n \mid v)$
  - How many data do we need?

- Let's add an assumption (a strong assumption)

$$P(a_1, a_2, \cdots, a_n \mid v) = \prod_i P(a_i \mid v)$$

If Output is given,
Inputs are independent from each other.

Formal Description of NBC

- Naïve Bayesian Assumption

$$P(a_1, a_2, \cdots, a_n \mid v) = \prod_i P(a_i \mid v)$$

If Output is given,
Inputs are independent from each other.

  - Acceptable? Why?
  - Helpful? How many data are necessary to evaluation $P(a_i \mid v)$

Formal Description of NBC

- What's Next?
  - Estimate those based on given example

$$P(v) \text{ and } P(a_i) \text{ for } i = 1, \cdots, n$$

  - Putting all the probabilities and choose v

$$f(x) = \arg\max_{v \in V} P(v) \prod_i P(a_i \mid v)$$

Performance

- Conditional independence assumption is often violated but it is easy to implement and works surprisingly well anyway

- When to use:
  - Moderate or large training set available
  - Attributes that describe instances are conditionally independent given classification
- Successful applications:
  - Diagnosis
  - Classifying text documents: spam or non-spam

Discussion

- Let's do it once more with the following data

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
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4	Rain	Cool	Normal	Strong	No
5	Sunny	Mild	High	Weak	No
6	Rain	Mild	Normal	Weak	Yes
7	Overcast	Mild	High	Strong	Yes
8	Rain	Mild	High	Strong	No

- Today is Sunny, Mild, High and Strong
  - Will he play tennis today?

Discussion

- Naïve Bayesian Classifier
  - Let's ESTIMATE the followings, but HOW?

$$P(yes) = 3/8$$
$$P(no) = 5/8$$
$$P(sunny \mid yes) = 0/3$$
$$P(mild \mid yes) = 3/3$$
$$P(high \mid yes) = 2/3$$
$$P(strong \mid yes) = 1/3$$
$$P(sunny \mid no) = 3/5$$
$$P(mild \mid no) = 2/5$$
$$P(high \mid no) = 4/5$$
$$P(strong \mid no) = 3/5$$

→ ??

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
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5	Sunny	Mild	High	Weak	No
6	Rain	Mild	Normal	Weak	Yes
7	Overcast	Mild	High	Strong	Yes
8	Rain	Mild	High	Strong	No

Discussion

- Naïve Bayesian Classifier

$$P(yes \mid sunny, mild, high, strong)$$
$$= \frac{P(yes)P(sunny \mid yes)P(mild \mid yes)P(high \mid yes)P(strong \mid yes)}{P(sunny, mild, high, strong)}$$
$$= \frac{1}{\alpha}\left(\frac{3}{8} \times \frac{0}{3} \times \frac{3}{3} \times \frac{2}{3} \times \frac{1}{3}\right) = 0$$

$$P(no \mid sunny, mild, high, strong)$$
$$= \frac{P(no)P(sunny \mid no)P(mild \mid no)P(high \mid no)P(strong \mid no)}{P(sunny, mild, high, strong)}$$
$$= \frac{1}{\alpha}\left(\frac{5}{8} \times \frac{3}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{3}{5}\right) = \frac{0.072}{\alpha}$$

Is it OK?

Discussion

- If the size of given data or the number of instances for a specific value is small, NBC gives a poor performance

- When $n_{a_i,v}$ is very small:

$$P(a_i \mid v_j) = \frac{n_{a_i,v_j} + m \cdot p}{n_{v_j} + m}$$

  - n_v : number of training examples for which $v=v_j$
  - $n_{a_i,v}$: number of examples for which $v=v_j$ and $a=a_i$
  - p is a prior estimate
  - m is weight given to prior